Sahaj Bhadja

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Summary

Full-stack Software Engineer with 2+ years building **Python** and **JavaScript** applications, **JSON REST APIs**, and cloud data pipelines, with **test-driven development**, CI/CD, and healthcare data experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python/FastAPI** services and **SQL** schemas for 500+ users, adding validation and reconciliation checks across 5K+ monthly pipeline jobs while holding **99%+ reliability** across recurring production releases.
- Restructured a fault-tolerant async job orchestrator with explicit state transitions and failure isolation, processing 1K+ jobs and tripling throughput while cutting dashboard latency from 1.2s to 350ms.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **full-stack** features across a **React** front end, **Flask REST APIs**, and MySQL on **AWS** RDS, shipping through GitHub Actions **CI/CD** for 150+ instructors and 10K+ daily requests.
- Built a concurrent batch-processing engine with multithreading, secure file ingestion, and fault-tolerant execution, processing 1K+ files per batch and cutting feedback turnaround by 60% under peak load.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Maintained 20+ **unit and integration test** suites with **PyTest**, Mockito, and Jenkins **CI**, running nightly across three operating systems, catching regressions and improving pipeline efficiency by 68%.
- Troubleshot large-scale **API integrations** and redesigned **ETL** workflows on **AWS** Glue and S3 in **Agile** sprints with Microsoft/GitHub stakeholders, improving response times by 30% and cutting technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a MERN platform with **RESTful APIs**, WebSockets, and Azure SQL across 8 microservices, syncing 2M+ records with role-based access control and JWT authentication for six workflows.
- Automated Jenkins **CI/CD** and Docker packaging with staged approvals, cutting deployment time 50% while Sentry and Azure Monitor surfaced production issues for rapid triage across release cycles.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Gathered requirements and presented the system architecture to the client, mapping 12+ integration points across APIs, payments, and data sync before implementation, reducing design errors by 25%.
- Built Flask APIs with Firebase real-time synchronization and transaction locking to prevent double-booking of shared time slots, sustaining sub-200ms responses across 50K+ records on Android, iOS, and web.

Skills

Languages: Python, JavaScript/TypeScript, SQL, Java, C++, Bash
Full Stack: FastAPI, Flask, Django, React.js, Node.js, JSON REST APIs, HTML/CSS
Data & Storage: PostgreSQL, SQL Server, MySQL, SQLite, MongoDB, ETL Pipelines, Data Validation, Audit Trails
Testing & Domain: PyTest, Unit/Integration/End-to-End Testing, Test-Driven Development, Code Reviews, Healthcare Data, VCF/Variant QC, PHI Tokenization
Cloud & DevOps: AWS (EC2, S3, RDS, Glue), Azure, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Agile/Scrum

Projects

SpecimenFlow | *Python, FastAPI, SQLite, JavaScript, PyTest*

- Built a **LIMS** for a hereditary cancer panel in **Python/FastAPI**: specimen state machine, chain-of-custody audit, **PHI** tokenization, **VCF** parsing with **QC** filters, and a **JavaScript** dashboard, covered by **40 tests**.

HMIS Live | *MERN, ABDM APIs, TypeScript, Sentry*

- Built a healthcare platform integrating national health-record **APIs** with consent-based access to patient histories, Sentry monitoring, and automated SMS alerts across 100+ daily appointment workflows, reducing administrative workload by 30%.

Big-Brain Live | *Python, FAISS, PostgreSQL, GCP*

- Built a cloud document-analysis platform with embedding-based semantic search over 50K+ documents on **PostgreSQL** and GCP, cutting document review time by 60% for 200+ users while sustaining 95% query accuracy.